Air cushion (10 points)

Part A. Equilibrium of the platform

A.1	0.5pt

$c_V =$ _____

$c_P =$ _____

Answer: _____

A.2	0.5pt

$P_0/P_{atm} =$ _____

Answer: _____

Part B. How an air cushion works

B.1	0.5pt

$v =$ _____

Answer: _____

B.2	1.0pt

$P - P_0 =$ _____

Answer: _____

B.3	0.5pt

$c =$ _____

Answer: _____

B.4	0.5pt

$c_0 =$ _____

Answer: _____

B.5	1.0pt

$\mathrm{d}m/\mathrm{d}t =$ _____

Answer: _____

B.6	0.5pt

$v =$ _____

Answer: _____

B.7	1.5pt

$T/T_0 =$ _____

Answer: _____

Part C. Critical flow

C.1	1.0pt

$\mathrm{d}m/\mathrm{d}t =$ _____

$M =$ _____

(sketch on the working sheets)

Answer: _____

C.2	1.0pt

$\varepsilon =$ _____

$T =$ _____

$P =$ _____

$v =$ _____

$\rho =$ _____

Answer: _____

Part D. Uneven ground

D.1	1.5pt

$M =$ _____

$T =$ _____

$P =$ _____

$v =$ _____

$\rho =$ _____

Answer: _____